

# Combinatorial Game Theory

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## 1 Introduction

### Definition 1.1: Combinatorial Game Theory (CGT)

The study of finite 2-player, alternating turns, deterministic rules/moves, perfect information games.

### Definition 1.2: Normal Play Games

Whoever cannot move loses.

We can represent combinatorial games with a tree of all continuations. This is a directed acyclic graph where each node contains the game state and the player to move.

Games are played with **Left** and **Right**. In the tree, solid lines represent L moving and dotted represents R moving.

### Definition 1.3: ptm

Player-to-move.

### Definition 1.4: optm

Opponent-to-move.

### Definition 1.5: WLD Game

A combinatorial game that can result in a win, loss, or draw.

### Theorem 1.6 (Zermelo)

In a Win-Loss-Draw game, one of the following holds:

- ptm (first player) has a winning strategy.
- optm (second player) has a winning strategy.
- ptm and optm each have a draw strategy (a non-losing strategy).

**Proof.** Consider a Tree of all Continuations (ToaC). We prove this by induction. Assume the theorem holds for all continuations from the root  $g_1, \dots, g_t$ . We label each node of the each subtree with  $+$ ,  $-$  or  $0$ . There are three cases:

Case 1 At least 1 of the  $g_i$  is labeled  $-$ , say  $g_j$ .  $g_j = -$  means the player-to-move at  $g_j$  loses. Thus,

if you are at  $g$  and you move to  $g_j$ , then opponent is the player-to-move at  $g_j$  and loses. You force a win, so ptm wins.

Cbse 2 Assume not case 1 and that some  $g_j = 0$ . All  $g_i$ 's are 0 or +. You move to the  $g_j = 0$  and therefore guarantee a ptm-draw. If opponent moves first, then they can either move to a + and we win, or they move to 0 and forces a draw. So optm-draw under optimal play.

Ccse 3 Every  $g_i = +$ . Every move ptm makes goes to +, so next player can always win. Thus, optm-win.

■

## 1.1 Outcome Classes

### Definition 1.7: $\mathcal{Y}$

Left wins, no matter if they are 1st or 2nd player.

### Definition 1.8: $\mathcal{R}$

Right wins, no matter if they are 1st or 2nd player.

### Definition 1.9: $\mathcal{P}$

The 2nd player wins (0 game).

### Definition 1.10: $\mathcal{N}$

The 1st player wins.

## 2 Clobber

Clobber is a normal play combinatorial game. Each player either controls X or O and each turn they can capture an adjacent opponent symbol. The player who cannot move loses.

Linear clobber is when the game is played on a path graph. We can represent this in options notation:

- $x = \{\} = 0$ . The game tree is a single root node.
- $xo = \{0|0\} = *$ .
- $rox = \{0, 0|0, 0\} = \{0, 0\} = *$ .
- $xro = \{0|*\} = \uparrow$ .
- $oor = \{*|0\} = \downarrow$ .
- $xoro = \{0, *, xro|0, *, oor\}$ .

**Definition 2.1: Equivalent Games**

Two games  $G, H$  are equivalent if they have the same canonical form.

**Definition 2.2: Sum of Games**

For games  $A, B$ ,

$$A + B = \{A^L + B, A + B^L | A^R + B, A + B^R\}$$

for all Left options  $A^L, B^L$  and Right options  $A^R, B^R$ .

**Theorem 2.3 (Alternate Equivalent Games)**

Two combinatorial games  $G, H$  are equivalent iff for all games  $X$ ,  $\text{oc}(G + X) = \text{oc}(H + X)$  (outcome class).

The negative clobber game swaps each  $x$  to  $o$  and  $o$  to  $x$ .

**Definition 2.4: Negative Game  $-G$** 

If  $G = \{G^L | G^R\}$ , then  $-G = \{-G^R | -G^L\}$ .

E.g.  $G = \{xxo, xo, xxo | oox, ox\}$ , then

$$-G = \{xxo, xo | oox, ox, ooox\}$$

**Theorem 2.5 (Alternate Equivalent Games 2)**

Two combinatorial games  $G, H$  are equivalent if  $G + -H \in \mathcal{P}$ .

**Proposition 2.6**

$$\downarrow + * = \downarrow * \equiv \{0 | *, 0\}.$$

**Proof.** We have  $\downarrow = \{* | 0\}$  and  $* = \{0 | 0\}$ . By sums of games definition,

$$\downarrow + * = \{* + *, \downarrow + 0 | 0 + *, \downarrow + 0\} = \{0, \downarrow | *, \downarrow\}$$

Define  $G = \{0 | *, 0\}$  and  $H = \downarrow + * = \{0, \downarrow | *, \downarrow\}$ . To prove  $G \equiv H$ , we need to show  $G + -H \in \mathcal{P}$ .

First,  $-H = \{*, \uparrow | 0, \uparrow\}$ . We can compute  $G + -H$ , but there are two cases:

1. L plays first on  $G + -H$ , show R can win. If  $G$  plays first, then the three options of the tree are  $-H, G + *, G + \uparrow$ .  
 If  $-H$ , R wins using 0.  
 If  $G + *$ , then R can win by  $* + * = 0$ .  
 If  $G + \uparrow$ , then R plays to  $G + *$ . L plays to either  $*$  or  $G$ , which then R can both play to 0.
2. R plays first on  $G + -H$ , show L can win.

### Proposition 2.7

For all  $t \geq 2$ ,  $t \cdot \uparrow \equiv \{0|(t-1) \uparrow * \} > 0$  and  $t \cdot \uparrow * \equiv \{0|(t-1) \uparrow \}$ .

## 3 Hex

A combinatorial game played on a grid of hexagons. The goal is for one player to connect their colored sides using their same symbol/rock.

Hex has no draws.

### Theorem 3.1

For  $n \times n$  hex, the first player always wins.

**Proof.** We prove by contradiction. Let Left make the first play of the game, and assume that Right has a winning strategy. With Left's first move she puts a stone on any cell, a placement called "extra". With this move Left becomes Second, and she henceforth follows the winning strategy that is available to Right. At some point, if this strategy calls for Left to place a stone where the extra sits; then she will simply make another arbitrary placement. Thus Left can win in contradiction to the hypotheses. ■

## 4 Go

Go is played on a graph, typically a grid. It is a combinatorial game where the winning condition is more stones and territory occupied. A legal move must be played on an empty cell and the block (a connected component) must have at least 1 liberty/free space.

## 5 Nim

The game of Nim involves piles of items and on each turn, a player picks at least 1 item from a pile. The last person to pick items is the winner.

### Theorem 5.1 (Bouton)

There is a winning move iff the XOR sum is nonzero.

Consider the piles 6, 3, 1. In binary this is 110,011,001. The XOR sum is 100, thus there is a winning move for ptm.

There is a legal move from each pile whose leftmost bit matches the leftmost 1 bit of XOR sum. For the example, we add 100 to 110 to get 010, so we pick 2 from the first pile of 6 which leaves 4.

**Definition 5.2:  $*k$** 

The game of Nimber (Nim with one pile) with  $k$  stones.

**Definition 5.3: Impartial Game**

A game where both Left and Right have the same move options.

**Theorem 5.4**

Every impartial game is equivalent to a Nimber (one pile Nim).

**Theorem 5.5 (Nimsum)**

$$\sum_j *p_j \equiv *(\oplus p_j)$$

**Theorem 5.6 (Sprague-Grundy)**

Every impartial game is equivalent to a game of Nimber.

**Theorem 5.7 (MEX)**

For a game  $\alpha = \{*\alpha_1, \dots, *\alpha_k\}$ ,

$$\alpha \equiv * \text{mex}(\alpha_1, \dots, \alpha_k)$$

## 6 Domineering

**Definition 6.1: Cram**

A combinatorial, normal play game played on a grid of dots. On each move, you can cover 2 horizontally or vertically adjacent dots. Player who cannot make a move loses.

**Definition 6.2: Domineering**

A game of Cram, but Left can only play vertically and Right can only play horizontally.

The game  $1 = \{0|\}$  cannot be realized as a clobber game since Left only has one move.

Similarly,  $-1 = \{|\}0$ .

**Proposition 6.3**

$$G + 1 \in \mathcal{L}, -1 + G \in \mathcal{R}$$

**Proof.** First  $G + 1 \in \mathcal{L}$ . There are two cases:

1. Assume Left is player 1. Play on  $G$ . If result  $G \rightarrow 0$ , so total game is  $1 + 0 = 1$ . It is Right's turn and Right has no moves by definition of 1 game.

If result is  $G \rightarrow G'$ , so total game is  $G' + 1$ . It is now Right's turn and Right must play on  $G'$  since Right has no moves in 1. So on Right's turn, so  $G' + 1 \rightarrow 0$  or  $G' + 1 \rightarrow G'' + 1$ . For the 0 case, Left can play on 1 and wins. For the second case, continue by induction on depth of tree.

2. Assume Right is player 1. Same as above, when Right plays on  $G + 1$ .

■

#### Proposition 6.4

A smallest game  $g$  such that  $g \equiv 3 \times 3$  domineering game is  $g = \pm 1 = \{1 | -1\}$ .

**Proof.** There are two cases:

1.  $g \equiv \pm 1$ : We need to show  $g + -\pm 1 \in \mathcal{P}$ . There are two cases:
  - Left plays first on  $g + -\pm 1$ , show Right wins.  $-\pm 1 = \pm 1$  since you invert and negate. So we show  $g + \pm 1 \in \mathcal{P}$ .
  - Right plays first on  $g + -\pm 1$ , show Left wins.

We would do this for more complicated proofs, but we can notice  $g = \pm 1$ .

2. There is no smaller game (tree depth): The game tree  $g$  has depth 2. We show that  $g \not\equiv$  any game with depth 0 or 1.  $g$  is in  $\mathcal{N}$ , but 0 is in  $\mathcal{P}$ , so it is not equivalent to a depth 0 game. For depth 1, ...

■

## 7 Canonical Form

#### Definition 7.1: Canonical Form

The minimally equivalent game.

#### Definition 7.2: $A < B$

$A < B$  if  $B + -A \in \mathcal{L}$ .

#### Definition 7.3: $A \leq B$

$A \leq B$  if  $A \equiv B$  or  $A < B$ .

**Theorem 7.4 (Canonical Form I [Conway, Berlekamp, Guy])**

If  $G = \{L_1, L_2, \dots | R_1, R_2, \dots\}$  and  $L_1 \leq L_2$ , then

$$G \equiv \{L_2, \dots | R_1, R_2, \dots\}$$

If  $G = \{L_1, L_2, \dots | R_1, R_2, \dots\}$  and  $R_1 \geq R_2$ , then

$$G \equiv \{L_1, L_2, \dots | R_2, \dots\}$$

This shows to prune inferior options.

**Proposition 7.5**

$$0 < 1$$

**Proposition 7.6**

$$-1 < 1$$

**Definition 7.7:  $A > B$** 

$A > B$  if  $B + -A \in \mathcal{R}$ .

For example,  $G = \{0, 1 | 1, -1\}$ , we can prune 0 from the left options since  $0 < 1$ , so  $G \equiv \{1 | 1, -1\}$ . And also,  $1 > -1$ , so we can prune 1 from the right options to get  $G \equiv \{1, -1\}$ .

**Definition 7.8: Fuzzy**

A fuzzy game is a game in outcome class  $\mathcal{N}$ .

**Definition 7.9: Reversible**

For a game  $g$ , a left (right) option  $L_1$  ( $R_1$ ) is reversible if it has right (left) option  $LR_1$  ( $RL_1$ ) such that  $LR_1$  ( $RL_1$ ) is at least as good for Right (Left) as  $g$

Let  $g = \{0, * | *\}$ . Say  $L_1$  here is  $*$  and  $LR_1$  is 0. Does Right prefer  $LR_1$  to  $g$ , i.e.  $LR_1 = 0 \leq g$ ?

Either  $g \equiv 0$  leads to position  $\mathcal{P}$  or  $g > 0$  is in  $\mathcal{R}$ .

Yes, we can replace  $*$  for left options to the game  $\{0 | *\}$ .

The above is  $g$  is Right-preferable to  $h$  if  $g \leq h$ .  $g$  is Left-preferable to  $h$  if  $g \geq h$ .

**Theorem 7.10 (Canonical Form II – Reversible)**

If  $LR_1$  is reversible, then we can replace  $L_1$  with set of left options of  $LR_1$ .

**Proposition 7.11**

$$1 \leq *$$

**Proof.** Consider  $1 - *$ , we show that Left has a 2-player win strategy. So we only need to consider Right first moves. Right moves to 1, and Left wins. ■

**Definition 7.12:**  $\frac{1}{2}$

$$\frac{1}{2} = \{0|1\}$$

The game  $\frac{1}{2}$  is the same as the domineering game that looks like an L block since Left can play to either 0 or  $-1$ , and Right can play in the one spot, so it plays to 1.

Claim:  $g + g + -1 \in \mathcal{P}$ . This  $g$  is called  $\frac{1}{2}$  from this.

## 7.1 Counting Minimal Games

The question here is to ask how many games  $g$  are there that  $CF(g)$  has depth  $\leq k$ .

For depth  $\leq 0$ , there is exactly one game with depth 0, the game 0.

For depth  $= 1$ , there are 3 games  $1, -1, *$ . So there are at most 4 games with depth  $\leq 1$ .

For depth  $\leq 2$ , there are at most  $2^4 \times 2^4 = 256$  games. This is because there are 4 games of depth  $\leq 1$ . There are  $2^4$  sets of left options.

Since the move options are dominant, there are only really 6 possible Left and 6 possible Right move options. Careful analysis shows there are only 22 unique games.

Table 1: Canonical form of games with depth  $\leq 2$

| L/R         | $\emptyset$ | 0              | -1            | 1             | *          | $0, *$         |
|-------------|-------------|----------------|---------------|---------------|------------|----------------|
| $\emptyset$ | 0           | -1             | -2            | 0             | 0          | -1             |
| 0           | 1           | *              | $\{0 -1\}$    | $\frac{1}{2}$ | $\uparrow$ | $\downarrow *$ |
| 1           | 2           | $\{1 0\}$      | $\pm 1$       | $1*$          | $\{1 *\}$  | $\{1 0, *\}$   |
| -1          | 0           | $-\frac{1}{2}$ | $-1*$         | 0             | 0          | $-\frac{1}{2}$ |
| *           | 0           | $\downarrow$   | $\{*-1\}$     | 0             | 0          | $\downarrow$   |
| $0, *$      | 1           | $\uparrow *$   | $\{0, * -1\}$ | $\frac{1}{2}$ | $\uparrow$ | $*2$           |

## 8 Numbers

**Definition 8.1: Number**

A normal-play, combinatorial game with no loops but possibly infinite.

**Definition 8.2: Dyadic Rationals**

A normal-play, combinatorial game with no loops (finite).



**Definition 8.3: Reals**

A normal-play, combinatorial game with no loops (possibly infinite).

**Definition 8.4: Surreals**

A normal-play, combinatorial game with no loops (infinite).

**Definition 8.5: Number (Recursive)**

A game  $g$  where  $\forall g^L$  is a number and  $\forall g^R$  is a number, and no  $g^L \geq g^R$ .

Here  $\forall g^L$  means for each individual left option.

E.g.  $0 = \{|\}$  is a number.  $\{0|\}$ ,  $\{|\}$ ,  $\{0|1\}$  are numbers, but  $\{1|0\}$  is not a number.

$\{1, 2|4, 5\}$  is a number. By CF 1, this is equivalent to  $\{2|4\}$ .

$\{*|\}$  is not a true number by the definition, but is equivalent to the game 0, which is a number.

**Theorem 8.6**

Numbers are not fuzzy.

**Proof.** Let  $g = \{g^L|g^R\}$  is a number. Assume  $g||0$ . Then there is some  $g^L \geq 0$  (Left prefers larger numbers) and some  $g^R \leq 0$  (Right prefers smaller numbers). But  $g^L \geq 0 \geq g^R$ , which contradicts the definition of a number. ■

**Theorem 8.7**

Let  $g = \{g^L|g^R\}$  be a number and  $\forall h \in g^L, \forall k \in g^R$ , then

$$h < g < k$$

Let a game of Hackenbush with one blue line on the ground and two red lines vertically on top of the blue line. This game is

$$\left\{0|\frac{1}{2}, 1\right\} = \left\{0|\frac{1}{2}\right\} = \frac{1}{4}$$

You can prove for all Hackenbush games  $g$ ,  $\forall g^L < g$  and  $\forall g^R > g$ , then  $g$  is a number.

You can increase the number of red lines to get the sequence  $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \dots$

**Theorem 8.8**

For all  $n \geq 1$ ,

$$\frac{1}{2^n} = \left\{0|\frac{1}{2^{n-1}}\right\}$$

We can create a binary search tree of all numbers. A number of on depth  $t$  if its canonical form has depth  $t$ .

### Theorem 8.9 (Simplicity Principle for Dyadic Games)

Let  $g = \{\bullet\alpha_1, \dots, \bullet\alpha_n \mid \bullet\beta_1, \dots, \bullet\beta_m\}$  with  $\alpha_j < \beta_k$  for all  $j, k$ , then  $g \equiv \bullet\gamma$ , where  $\gamma$  is the simplest number between  $\max(\bullet\alpha_j)$  and  $\min(\bullet\beta_k)$ .

This is similar to the MEX theorem.

### Theorem 8.10

A dyadic rational can be represented as a Hackenbush game, and vice versa.

## 9 Zero-Sum Matrix Games

We move away to classical game theory.

### Definition 9.1: Zero-Sum Matrix Game

A game played between P1 and P2 and there is a matrix, where if they play to  $(i, j)$ , then  $A[i, j]$  is the payoff of the game.

For P1 in Rock-Paper-Scissors, where P1 are the columns and P2 are the rows. The payoff matrix is

$$\begin{bmatrix} 0 & 1 & -1 \\ -1 & 0 & 1 \\ 1 & -1 & 0 \end{bmatrix}$$

Worst-case analysis just shows that we always lose, i.e.  $-1$ . A better way to analyze is through average-case analysis. This is essentially a probability distribution analysis.

Assume P1 and P2 plays uniformly at random. Then each row and column has a chance of  $\frac{1}{3}$ . Thus, each entry of the matrix has a  $\frac{1}{9}$  chance of being played. We attach a  $+$  or  $-$  depending if P1 or P2 wins.

$$\begin{bmatrix} - & 1/9 & -1/9 \\ -1/9 & - & 1/9 \\ 1/9 & -1/9 & - \end{bmatrix}$$

The expected win score for P1 is the sum of entries  $= 0$ . Even for different probabilities, we will still get expected score of 0.

### Definition 9.2: Strategy

A vector for one of the players that will tell you how much of each row or column the player selects.

### Definition 9.3: Pure Strategy

The vector where one entry is 1 and the others are 0. This represents choosing exactly one of the rows or columns.

**Definition 9.4: Mixed Strategy**

A vector where each entry is in  $(0, 1)$  and the sum of entries is 1.

Given column strategy  $c$  and row strategy  $r$  for matrix  $A$ , the expected payoff is  $r^T A c$ .

The dominated options is similar to combinatorial games. Rose prefers if all entries in a row if they are *larger* than the other rows. Similarly, Colin prefers entries in columns that are *smaller* than the other columns.

## 9.1 Using LP

Observe that given any mixed strategy, there exists a best counter mixed strategy that is pure.

Thus, given a distribution  $r$  for the Rows, we only need to consider all pure strategies to get functions of  $r$ . Then Rows wants to maximize the function  $\min(\text{functions}(r))$ .

We can do this with an LP by introducing a new variable  $z$ , where the objective function is  $\max z$  subject to  $z \leq \text{functions}(r)$ ,  $r_1 + \cdots + r_m = 1$ , and  $r \geq 0$ .